

MSML642 Homework 2

Time Series Prediction with RNN and LSTM

In this assignment, you will build and compare a vanilla recurrent neural network and a long short-term memory model for one-step time-series prediction. You will use the same data split and evaluation process for both models so that the comparison is meaningful.

Learning objectives

- Prepare sequential data for supervised learning without leaking test information into training.
- Implement vanilla RNN and LSTM regression models in PyTorch.
- Evaluate both models on unseen data and explain the observed differences.

Dataset

Use the Airline Passengers dataset:

<https://raw.githubusercontent.com/jbrownlee/Datasets/master/airline-passengers.csv>. The target is the monthly number of international airline passengers, reported in thousands. Treat the observations as a chronological sequence.

Required work

1 Data loading visualization and preprocessing (20 points)

- Load the dataset and confirm that the month and passenger columns are usable.
- Check the data type, shape, and missing values.
- Plot passenger counts over time with a title and labeled axes.
- Convert the target data to a numeric format suitable for PyTorch.

2 Chronological split and sequence creation (15 points)

- Choose and report an input sequence length.
- Split the series chronologically into training and test sets. Do not randomly shuffle observations before the split.
- Fit any scaler only on the training data, then apply it to both sets.
- Create input and target tensors with the shapes required by PyTorch sequence models.

3 Vanilla RNN model (40 points)

- Implement a regression model with `torch.nn.RNN` and a final output layer.
- Document the input size, hidden size, number of layers, and output shape.

4 LSTM model (40 points)

- Implement a regression model with `torch.nn.LSTM` and a final output layer.
- Use the same data split and sequence length as the RNN model.

5 Training evaluation and comparison (30 points)

- Train both models and report the selected optimizer, learning rate, number of epochs, and loss function.
- Evaluate both models on the test set using MAE and RMSE.
- Plot the test predictions from both models against the ground truth.
- Compare the results and explain which model performed better, what the metrics show, and where the models make errors.

6 Code organization and explanations (5 points)

- Organize the notebook so it runs from top to bottom after all TODO items are completed.
- Use Markdown cells to explain preprocessing choices, model settings, results, and conclusions.

Optional bonus robotics time series extension

Up to 15 bonus points. Complete one additional time-series prediction task using one of the robotics-related datasets below, or another publicly available robotics-related dataset. Submit the bonus work in the same notebook and exported PDF as the required assignment.

Bonus requirements

- Identify the dataset and source. State the input features, prediction target, and sequence length.
- Load and inspect the data, then include a labeled plot of the target over time or across sequence steps.
- Split continuous data chronologically. For separate trials, routes, or engines, split whole sequences so one run cannot appear in both training and test sets. Fit any scaler on training data only.
- Train either an RNN or LSTM. State the model settings, optimizer, learning rate, epochs, and loss function.
- Report test MAE and RMSE in the target's original units, and plot test predictions against ground truth.
- Explain how the prediction could be useful in a robotic or autonomous system, interpret the result, and note one limitation.

Suggested datasets

Dataset	Suggested prediction	Difficulty
UR3 CobotOps	Predict the next joint temperature from recent current, temperature, and speed measurements.	Easy
Robot Execution Failures	Predict the final force or torque sample from the preceding 14 samples.	Easy
RoboNav Robot Data	Predict the next velocity or position using one robot and one processed route CSV.	Medium
UTIAS MRCLAM	Predict the next x and y robot position from recent odometry or pose history.	Medium
NASA C MAPSS	Predict remaining useful life from recent multivariate sensor history.	Advanced

Official dataset links

- **UR3 CobotOps:** [Open official dataset page](#)
- **Robot Execution Failures:** [Open official dataset page](#)
- **RoboNav Robot Data:** [Open official dataset page](#)
- **UTIAS MRCLAM:** [Open official dataset page](#)
- **NASA C MAPSS:** [Download dataset](#)

Using the provided data files

The student package includes the airline CSV and a small, representative selection from each bonus source. The airline CSV is for required work; choose only one bonus dataset if you attempt the extension. Extract ZIP files before reading their contents. Fit any scaler only on the training portion.

- **UR3 CobotOps:** Extract `ur3_cobotops.zip` and read `dataset_02052023.xlsx` with `pandas.read_excel` (`openpyxl`). Sort by Timestamp; for example, predict the next `Temperature_J1` from recent current, speed, and temperature readings. Keep cycles separate when splitting.
- **Robot Execution Failures:** Extract `robot_execution_failures.zip`. Each block in `lp1.data` starts with a class label, then 15 rows of six force/torque values. Predict the 15th sample from the first 14; split whole blocks, never individual rows. Do not run the included `a.out` file.
- **RoboNav:** Read `robonav/husky_route1_processed.csv` with `pandas.read_csv(..., sep=';')`. For example, predict the next `navsat_vel` from the preceding 12 rows. Keep row order and split the route chronologically.

- UTIAS MRCLAM: Read mrclam/Robot1_Groundtruth.dat as whitespace-separated text with comment='#'. Its columns are time, x, y, and orientation. Predict the next x and y from recent poses; split by time. The package contains only Robot 1 from Dataset 3.
- NASA C MAPSS: Extract nasa_cmapss.zip and its nested CMAPSSData.zip; start with train_FD001.txt. The first two whitespace-separated columns are engine ID and cycle. Estimate remaining useful life from sensor history, and split by whole engine IDs to avoid leakage.

Robotics bonus grading

- Dataset, prediction target, and preprocessing: 3 points
- RNN or LSTM model: 4 points
- MAE, RMSE, and prediction plot: 4 points
- Robotics use, result interpretation, and limitation: 4 points

Grading

Section	Points
Data loading visualization and preprocessing	20
Chronological split and sequence creation	15
Vanilla RNN model	40
LSTM model	40
Training evaluation and comparison	30
Code organization and explanations	5
Base total	150
Optional robotics extension	+15
Maximum total with bonus	165

Submission requirements

- Submit MSML642_FirstName_LastName_HW2.ipynb.
- Export the completed notebook and submit MSML642_FirstName_LastName_HW2.pdf.
- Submit both files directly. Do not place them in a ZIP archive.
- Run the completed notebook from top to bottom before submission and keep the outputs visible.
- Cite external code or data and follow the course policy for collaboration and AI tools.
- Late submissions follow the course syllabus.